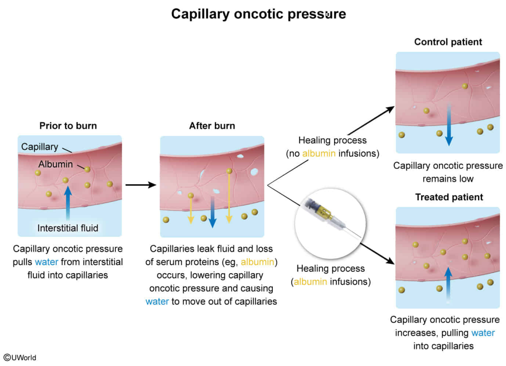
When body surface area is affected by a burn, capillaries leak fluid containing serum proteins such as albumin. A burn patient is treated with continuous intravenous infusions of albumin during the healing process. Compared to a burn patient control (no albumin infusions), the treated patient's capillary oncotic pressure will most likely be:

- A. increased. [63%]
- B. decreased. [21%]
- C. the same. [3%]
- D. increased, but fluid leakage will also increase over time. [11%]

Correct
63% answered correctly

Explanation:



During osmosis, water generally flows from *lower* to *higher* osmotic pressure environments across cell membranes. Osmotic pressure refers to the tendency of water to cross a semipermeable membrane by osmosis. This tendency is the result of solute concentration differences on either side of the membrane: A *higher* solute concentration creates a *higher* osmotic pressure. Osmosis occurs across capillary walls (ie, semipermeable membranes) into the interstitial fluid and vice versa due to differences in the concentrations of solutes (eg, plasma proteins such as albumin). This also occurs because of osmotic pressure, but in the capillaries, it is known as **oncotic pressure**.

Due to their size, plasma proteins are the only components of plasma that *cannot* easily move through pores in the capillary wall into the interstitial fluid. Therefore, plasma proteins are responsible for maintaining capillary oncotic pressure. If oncotic pressure is greater in the capillaries than in the interstitial fluid (ie, if the plasma contains a higher concentration of solutes than the interstitial fluid), there is a pulling force for water to be drawn into the capillaries via osmosis. Conversely, if oncotic pressure is lower in the capillaries than in the interstitial fluid, water will be drawn from the capillaries into the interstitial fluid. The capillary oncotic pressure is opposed by the hydrostatic pressure, or pushing force, of the blood against the capillary walls.

This question states that when body surface area is affected by a burn, fluid leaks from the capillaries. As a result, the plasma loses proteins (ie, solutes) such as albumin, lowering the total plasma solute concentration. This in turn lowers the oncotic pressure in the capillaries, creating *less* of a pulling force for water movement into the capillaries.

If such a burn patient is treated with continuous intravenous infusions of albumin during the healing process, the capillary solute concentration will increase, which will ultimately *increase* the capillary oncotic pressure. This pressure will promote water movement *into* the capillaries via osmosis.

Compared to a control patient who did not receive albumin infusions, the treated patient will likely display **increased capillary oncotic pressure** due to a *higher* albumin (ie, solute) concentration in the plasma **(Choices B and C)**.

(Choice D) Although the patient receiving the albumin infusions will show an increased capillary oncotic pressure, fluid leakage from capillaries will decrease (*not* increase) over time as tissue healing occurs.

Educational objective:
Due to their size, plasma proteins such as albumin contribute to the capillary oncotic pressure. Water movement occurs across the capillary wall toward the side with a greater oncotic pressure.

Time spent: 0 sec	**QID**:403083
Subject: Biology	**System:** Circulation and Respiration